

EXPERIMENT: 6

5G NR FRAME STRUCTURE ANALYSIS WITH DIFFERENT SUBCARRIER SPACING

Objective – 1: Analyze Different Subcarrier Spacing Values

MATLAB Program

```
clc;

clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

mu = 0:4; % Numerology Index

bar(scs)

grid on

set(gca,'XTick',1:5)

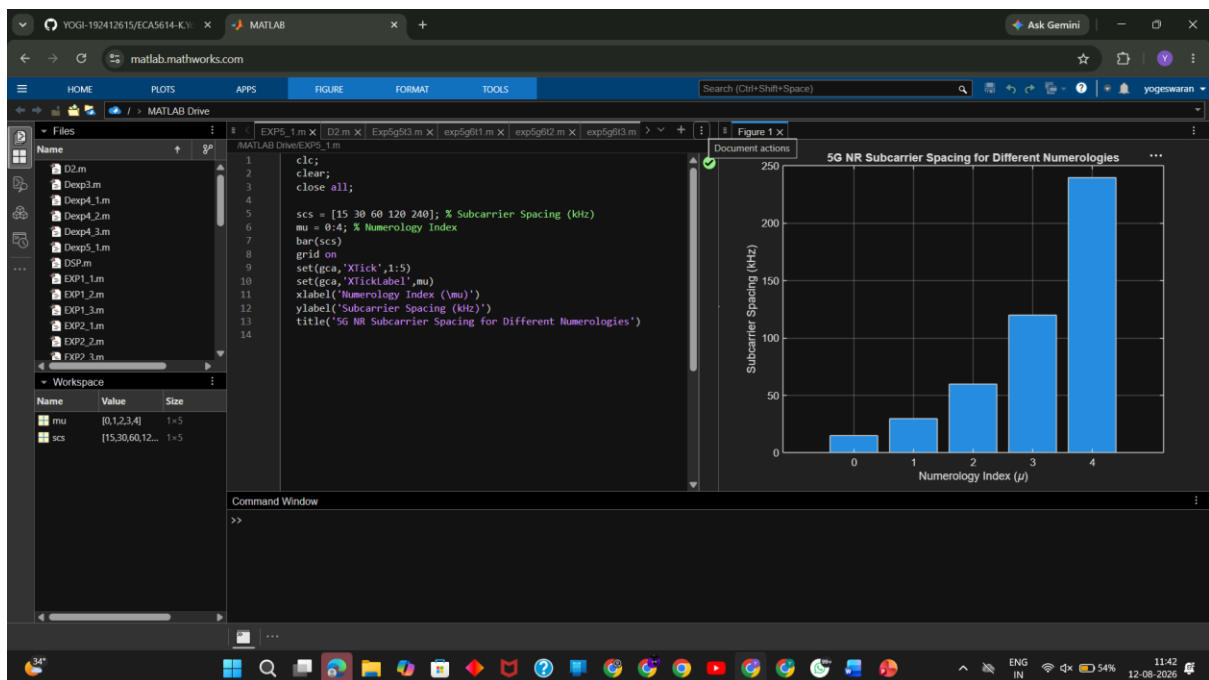
set(gca,'XTickLabel',mu)

xlabel('Numerology Index (\mu)')

ylabel('Subcarrier Spacing (kHz)')

title('5G NR Subcarrier Spacing for Different Numerologies')
```

OUTPUT



Objective – 2 Compare Slot Duration

Matlab Code :

```
clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

slot = [1 0.5 0.25 0.125 0.0625]; % Slot Duration (ms)

plot(scs, slot, 'o-', 'LineWidth', 2)

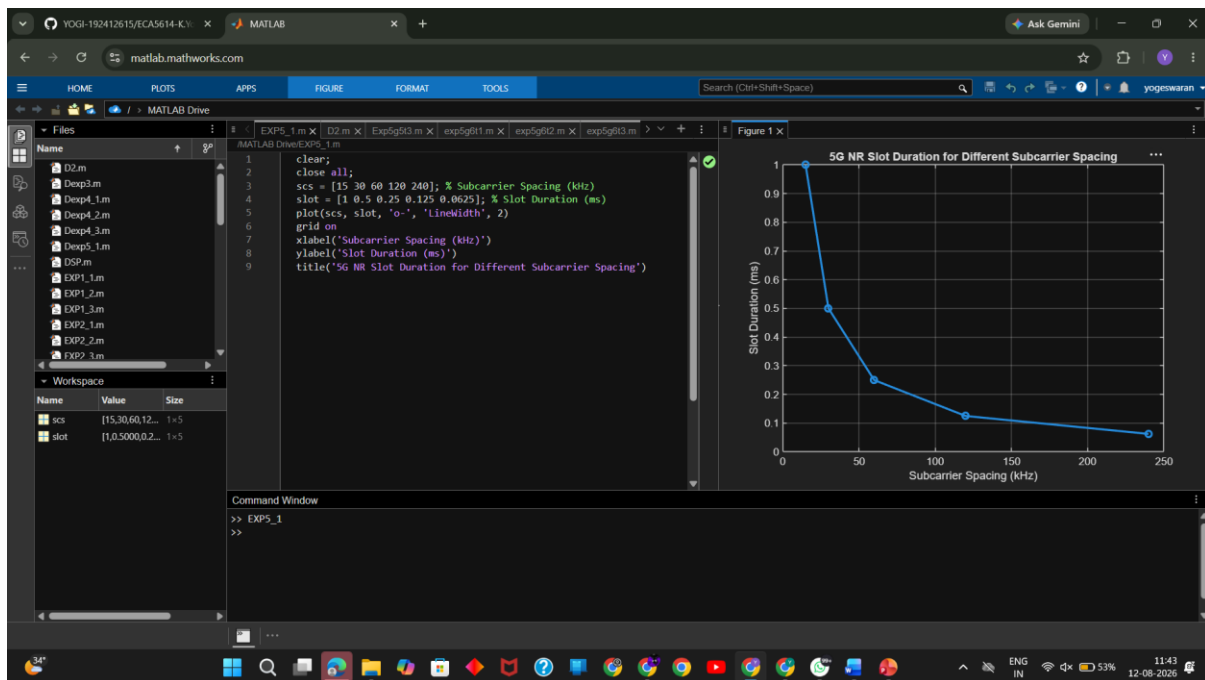
grid on

xlabel('Subcarrier Spacing (kHz)')

ylabel('Slot Duration (ms)')

title('5G NR Slot Duration for Different Subcarrier Spacing')
```

OUTPUT



Objective – 3 Study the Number of OFDM Symbols

Matlab Code

```
clc;

clear;

close all;

scs = [15 30 60 120 240]; % Subcarrier Spacing (kHz)

symbols = [14 14 14 14 14]; % OFDM Symbols per Slot

figure
```

```
bar(scs, symbols)
```

```
grid on
```

```
xlabel('Subcarrier Spacing (kHz)')
```

```
ylabel('Number of OFDM Symbols')
```

```
title('OFDM Symbols per Slot for Different Subcarrier Spacing')
```

OUTPUT

